

1. 闵可夫斯基

$$\sqrt{\sum (a_i + b_i)^2} \leq \sqrt{\sum a_i^2} + \sqrt{\sum b_i^2}$$

$$\sum (a_i + b_i)^2 = \sum a_i(a_i + b_i) + \sum b_i(a_i + b_i)$$

$$\leq \sqrt{\sum a_i^2} \sqrt{\sum (a_i + b_i)^2} + \sqrt{\sum b_i^2} \sqrt{\sum (a_i + b_i)^2}$$

$$= \sqrt{\sum (a_i + b_i)^2} (\sqrt{\sum a_i^2} + \sqrt{\sum b_i^2})$$

$$\Rightarrow \sqrt{\sum (a_i + b_i)^2} \leq \sqrt{\sum a_i^2} + \sqrt{\sum b_i^2}$$

2. 常见极限

$$x_n = \left(1 + \frac{1}{n}\right)^n \quad x_{n+1} = \left(1 + \frac{1}{n+1}\right)^{n+1}$$

$n \rightarrow +\infty$
 $n \rightarrow -\infty$ 都是 e

$$x_n = \underbrace{\left(1 + \frac{1}{n}\right) \cdot \dots \cdot \left(1 + \frac{1}{n}\right)}_{n \text{ 个}} \cdot 1 \leq \left(\frac{n(1 + \frac{1}{n}) + 1}{n+1}\right)^{n+1} = x_{n+1} \Rightarrow \text{单增}$$

$$\frac{1}{4} a_n = \frac{1}{2} \cdot \frac{1}{2} \cdot \underbrace{\left(1 + \frac{1}{n}\right) \cdot \dots \cdot \left(1 + \frac{1}{n}\right)}_{n \text{ 个}} \leq \left(\frac{2 \cdot \frac{1}{2} + n(1 + \frac{1}{n})}{n+2}\right)^{n+2} = 1 \Rightarrow a_n \text{ 有界}$$

$$a_n = \left(1 + \frac{1}{n}\right)^n = 1 + C_n^1 \frac{1}{n} + C_n^2 \left(\frac{1}{n}\right)^2 + \dots = 1 + 1 + \frac{1}{2!} \frac{n(n-1)}{n^2} + \frac{1}{3!} \frac{n(n-1)(n-2)}{n^3} + \dots$$

$$= 1 + \sum_{k=1}^n \left[\frac{1}{k!} \prod_{j=1}^{k-1} \left(1 - \frac{j}{n}\right) \right]$$

$$\leq 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} < 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + \dots$$

$$\text{要证 } \lim_{n \rightarrow \infty} 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!} = e$$

$$\text{固定 } m, \forall n > m \quad a_n = (1 + \frac{1}{n})^n > 1 + 1 + \frac{1}{2!}(1 - \frac{1}{n}) + \frac{1}{3!}(1 - \frac{1}{n})(1 - \frac{2}{n}) + \dots$$

$$e \geq S \quad \text{令 } n \rightarrow \infty \quad e = \lim_{n \rightarrow \infty} a_n \geq 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{m!} = b_m$$

$$\text{令 } m \rightarrow \infty \quad \lim_{m \rightarrow \infty} b_m \leq e$$

$$e \leq S \quad a_n \leq \sum_{k=0}^n \frac{1}{k!} \leq \sum_{k=0}^{\infty} \frac{1}{k!} = S$$

$$\star \lim_{n \rightarrow \infty} (1 - \frac{1}{n})^n = \lim_{n \rightarrow \infty} (\frac{n-1}{n})^n = \lim_{n \rightarrow \infty} \frac{1}{(1 + \frac{1}{n-1})^n} = \lim_{n \rightarrow \infty} \frac{1}{(1 + \frac{1}{n-1})^{n-1} (1 + \frac{1}{n-1})} = \frac{1}{e}$$

$$f(n) = (1 + \frac{1}{n})^n$$

$$\ln f(n) = n \ln \frac{n+1}{n} < n \cdot \frac{1}{n} = 1 \quad \Leftrightarrow \text{啥都没证.}$$

$$\text{证明: } 0 < e - (1 + \frac{1}{n})^n < \frac{3}{n} \quad (n=1, 2, \dots)$$

左显然.

右. 利用 $e < (1 + \frac{1}{n})^{n+1}$

$$\text{先证 } e < (1 + \frac{1}{n})^{n+1}$$

$$\Leftrightarrow 1 < (n+1) \ln \frac{n+1}{n}$$

$$\text{而 } \ln x > 1 - \frac{1}{x} \quad \text{证毕.}$$

$$\text{右} \Leftrightarrow \text{证 } (1 + \frac{1}{n})^{n+1} < \frac{3}{n} + (1 + \frac{1}{n})^n$$

$$\Leftrightarrow \text{证 } (1 + \frac{1}{n})^n < 3$$

取 $n=1$. 证毕.

$$\text{从 } \lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n \text{ 到 } \lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x$$

$$[x] \leq x < [x] + 1$$

$$1 + \frac{1}{[x]+1} < 1 + \frac{1}{x} \leq 1 + \frac{1}{[x]}$$

$$e - \varepsilon < (1 + \frac{1}{[x]+1})^{[x]} < (1 + \frac{1}{x})^x < (1 + \frac{1}{[x]})^{[x]+1} = (1 + \frac{1}{[x]})^{[x]} \cdot (1 + \frac{1}{[x]}) < e + \varepsilon$$

$$\forall \varepsilon > 0, \exists N_1, \text{ s.t. } x > N_1 \text{ 时 } (1 + \frac{1}{x})^x < e + \varepsilon$$

$$\exists N_2, \text{ s.t. } x > N_2 \text{ 时 } (1 + \frac{1}{x})^x > e - \varepsilon$$

$$\text{取 } \delta = \varepsilon \quad 0 < |x - x_0| < \delta \text{ 时 } \text{取 } N = \max\{N_1, N_2\}, \text{ 当 } x > N \text{ 时, } \left| (1 + \frac{1}{x})^x - (1 + \frac{1}{[x]+1})^{[x]} \right| < \varepsilon, \left| (1 + \frac{1}{x})^x - (1 + \frac{1}{[x]})^{[x]+1} \right| < \varepsilon.$$

$$\Rightarrow \lim_{x \rightarrow \infty} (1 + \frac{1}{x})^x = e$$

$$\lim_{x \rightarrow -\infty} (1 + \frac{1}{x})^x = \lim_{y \rightarrow +\infty} \frac{1}{(1 - \frac{1}{y})^y} = \lim_{y \rightarrow +\infty} (\frac{y-1+1}{y-1})^y = \lim_{t \rightarrow \infty} (t + \frac{1}{t})^{t+1} = e$$

$$\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$$

泰勒展开

$$f(x) = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots + \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x)$$

麦克劳林展开式

$$f(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \dots + \frac{f^{(n)}(0)}{n!}x^n + R_n(x)$$

$$\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e \quad \text{求证} \quad \lim_{n \rightarrow \infty} (1 + \frac{1}{2!} + \frac{1}{3!} + \dots + \frac{1}{n!}) = e$$

$$(1 + \frac{1}{n})^n = \sum_{k=0}^n \frac{1}{k!} (1 - \frac{1}{n})(1 - \frac{2}{n}) \dots (1 - \frac{k-1}{n})$$

对于每个固定的 k 当 $n \rightarrow \infty$, 乘积部分 $\rightarrow 1$

$$\Rightarrow (1 + \frac{1}{n})^n \rightarrow \sum_{k=0}^{\infty} \frac{1}{k!}$$

$\sum_{k=0}^n \frac{1}{k!}$ 升, 且 $\frac{1}{k!} \leq \frac{1}{2^{k-1}} \Rightarrow$ 收敛于极限 s

$$\prod_{j=0}^{k-1} (1 - \frac{j}{n}) \leq 1 \Rightarrow (1 + \frac{1}{n})^n \leq s_n \Rightarrow e \leq s$$

对固定 $m < n$ 有 $(1 + \frac{1}{n})^n \geq \sum_{k=0}^m \frac{1}{k!} (1 - \frac{1}{n})(1 - \frac{2}{n}) \dots (1 - \frac{k-1}{n})$

$$n \rightarrow \infty \text{ 时} \quad e \geq \sum_{k=0}^m \frac{1}{k!} \Rightarrow e \geq s$$

综上 $e = s$

详解

$\forall n$ 固定, 对 $\forall m > n$

$$a_m = (1 + \frac{1}{m})^m = \sum_{k=0}^m \frac{1}{k!} (1 - \frac{1}{m})(1 - \frac{2}{m}) \dots (1 - \frac{k-1}{m})$$

$$= \sum_{k=0}^n \dots + \sum_{k=n+1}^m \dots > \sum_{k=0}^n \frac{1}{k!} (1 - \frac{1}{m})(1 - \frac{2}{m}) \dots (1 - \frac{k-1}{m}) = t$$

舍掉, 使其成为有限项.

可用极限的四则运算.

$$m \rightarrow \infty \text{ 时, 对 } k \leq n \text{ 有 } \lim_{m \rightarrow \infty} (1 - \frac{1}{m})(1 - \frac{2}{m}) \dots (1 - \frac{k-1}{m}) = 1$$

$$\lim_{m \rightarrow \infty} a_m = e$$

$$\Rightarrow e \geq t$$

斐波那契数列相邻项比值

法1: 假设极限存在.

$$F_{n+1} = F_n + F_{n-1}$$

$$R_n = \frac{F_{n+1}}{F_n} = 1 + \frac{F_{n-1}}{F_n} = 1 + \frac{1}{R_{n-1}}$$

$$L = 1 + \frac{1}{L} \Rightarrow L = \frac{\sqrt{5}+1}{2}$$

$$\text{取 } A_n = R_{2n} \quad B_n = R_{2n+1}$$

$$A_1 = 2, A_2 = \frac{5}{3}, \dots$$

$$B_0 = 1, B_1 = \frac{3}{2}, \dots$$

$$A_n = R_{2n} = 1 + \frac{1}{R_{2n-1}} = 1 + \frac{1}{B_{n-1}}$$

$$B_n = R_{2n+1} = 1 + \frac{1}{R_{2n}} = 1 + \frac{1}{A_n}$$

$$A_1 = 2 > \varphi \quad \text{若 } A_{n-1} > \varphi = \frac{\sqrt{5}+1}{2}$$

$$A_n = \frac{2A_{n-1}+1}{A_{n-1}+1} = 2 - \frac{1}{A_{n-1}+1} > 2 - \frac{1}{\frac{3+\sqrt{5}}{2}} = \frac{\sqrt{5}+1}{2} = \varphi$$

$$\Rightarrow A_n = \frac{2A_{n-1}+1}{A_{n-1}+1} \Rightarrow A_n - A_{n-1} = -\frac{A_{n-1}+A_{n-1}+1}{A_{n-1}+1} < 0$$

$$\Rightarrow A_n \downarrow$$

$$\text{同理} \Rightarrow B_n \uparrow, B_n < \varphi$$

故取极限 φ

$$x_1 = 1 \quad x_{n+1} = \frac{1}{1+x_n}$$

法2: 假设 $\{x_n\}$ 收敛于 A , 则 $A = \frac{1}{1+A} \Rightarrow A = \frac{\sqrt{5}-1}{2} \quad (A^2 = 1-A)$

$$|x_{n+1} - A| = \left| \frac{1-A-x_n A}{1+x_n} \right| = A \left| \frac{A-x_n}{1+x_n} \right| < \underbrace{A}_{\text{关键. 递推构造}} |A-x_n| < A^2 |A-x_{n-1}| < A^n |A-x_1| = \left(\frac{\sqrt{5}-1}{2}\right)^n \left|\frac{\sqrt{5}-3}{2}\right|$$

$$\lim_{n \rightarrow \infty} |x_{n+1} - A| < \lim_{n \rightarrow \infty} \left(\frac{\sqrt{5}-1}{2}\right)^n \cdot \frac{3-\sqrt{5}}{2} = 0$$

$$\Rightarrow \forall \varepsilon > 0, \quad |x_n - A| < \varepsilon.$$

$\Rightarrow \{x_n\}$ 收敛于 A .

$$x_n = \frac{F_{n+1}}{F_n} = 1 + \frac{1}{x_{n-1}} \quad \lim_{n \rightarrow \infty} x_n = A, \quad A = \frac{\sqrt{5}+1}{2}$$

$$0 \leq |x_n - \frac{\sqrt{5}+1}{2}| = \left| \frac{1}{x_{n-1}} - \frac{\sqrt{5}-1}{2} \right| = \frac{\sqrt{5}-1}{2} \cdot \frac{1}{x_{n-1}} \left| \frac{\sqrt{5}+1}{2} - x_{n-1} \right| < \dots < \left(\frac{\sqrt{5}-1}{2}\right)^{n-1} \left| x_1 - \frac{\sqrt{5}+1}{2} \right| = \left(\frac{\sqrt{5}-1}{2}\right)^n \rightarrow 0$$

夹逼定理 $\lim_{n \rightarrow +\infty} |x_n - \frac{\sqrt{5}+1}{2}| = 0$

